Is everything a Niche model?

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Abstract: TODO

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Introduction

The Niche Model Hegemony: Establish how the Niche Model (and its topological derivatives like the Cascade model) has historically served as the structural prior or template for most network theory.

The Philosophy Gap: Contrast structural/topological models (which prioritise species-agnostic link distributions) with realised/dynamic models like ATN and ADBM (which prioritise biological mechanisms like bioenergetics and optimal foraging).

The Research Question: Do these different philosophical starting points (mechanical vs. statistical) lead to the same conclusions about community stability, or does the choice of template fundamentally bias our understanding of network dynamics?.

1 Methods

1.1 Network generation

We generated food webs using six established generative models and a maximum entropy (MaxEnt) null model, spanning structural (Niche, Cascade, Random, MaxEnt) and trait-based approaches (Latent Trait Model, Allometric Trophic Network, Allometric Diet Breadth Model). All simulations were implemented in Julia using FoodWebTools and custom extensions.

To isolate model effects from input variation, all trait-based models were parameterised using an identical species pool at each replicate. Species richness was fixed (S=15), and for each iteration we sampled (i) a target basal fraction (U(0.1,0.3)), (ii) metabolic classes consistent with this fraction, and (iii) body masses from class-specific log-uniform distributions. These inputs were shared across the Latent Trait, ATN, and ADBM models. Structural models were instead parameterised by a target connectance drawn from U(0.05, 0.3).

Networks were generated using a rejection-sampling procedure to enforce comparable emergent properties. A network was retained only if its realised basal fraction and connectance fell within the target ranges (0.1–0.3 and 0.05–0.3, respectively). For trait-based models, networks were accepted only when all three models successfully generated valid networks from the same species pool. We obtained 100 valid replicates per model, with an upper bound on sampling attempts to prevent non-termination. MaxEnt networks were constructed by sampling joint in- and out-degree distributions consistent with a target number of links, followed by entropy-maximising rewiring under degree constraints. These networks were subjected to the same acceptance criteria as other models. For each network, we recorded the adjacency matrix, model identity, input parameters, emergent structural properties, and (where applicable) species traits.

Table 1: Summary of food web models used in this study. Structural models (Niche, Cascade, MaxEnt) generate network topology based on abstract or statistical rules, while realised/dynamic models (ADBM, ATN, Latent trait) incorporate species-specific traits and biological mechanisms to predict interactions. For each model, we list the main inputs and the key assumptions that govern link formation.

Model	Type	Main Inputs	Key Assumptions
Niche model (Williams & Martinez, 2000)	Structural	Species niche values	Trophic interactions structured by a 1D feeding niche; species consume all prey within a contiguous range; allows cannibalism.
Cascade model (Cohen et al., 1990)	Structural	Species rank	Consumers feed only on species with lower ranks; strictly hierarchical; no loops or omnivory.
MaxEnt model (Banville et al., 2023)	Structural	Species richness, connectance	Links assigned to maximize entropy; captures global network properties without species-specific mechanisms.
Allometric Diet Breadth Model (ADBM) (Petchey et al., 2008)	Realised / Dynamic	Body mass, prey energy content	Consumers maximize energy intake; diet breadth determined by profitability and handling time; allometric scaling governs parameters.
Allometric Trophic Network (ATN) (Brose et al., 2006)	Realised / Dynamic	Body mass	Interactions constrained by body-mass ratios; mechanical size limits structure networks; probability of interaction follows Ricker function.
Latent trait model (Rohr et al., 2010)	Realised / Dynamic	Species traits (*e.g.,* body mass)	Interactions inferred statistically based on hidden trait correlations; can incorporate probabilistic constraints on links.

1.2 Network structure

All networks were converted to directed interaction networks and analysed using a standardised pipeline. We quantified structural properties across multiple scales, including connectance, trophic level, generality, vulnerability, clustering, centrality, trophic coherence, and path length. These metrics were used to characterise the structural signature of each model.

1.3 Dynamic simulations

To evaluate the dynamical consequences of different generative models, all networks were subjected to a common bioenergetic simulation framework using EcologicalNetworksDynamics.jl.

For trait-based models (ATN, ADBM, LTN), species body masses were taken from the generative netwroks and rescaled relative to the producer with the lowest body mass. These models were then parameterised with their corresponding body masses and stardard allometric scaling. For structual models (Niche, Cascade, MaxEnt, Random), where explicit trait information was not available, we assigned body masses using trophic scaling with a predator–prey body-mass ratio of 100, a value commonly used for aquatic food webs. Initial species biomasses were randomly assigned. Dynamics were simulated using a Type III functional response, with a Hill exponent of 2.

Each network was simulated until a steady state was reached, with a minimum simulation time of 2000 time units and a maximum of 5000 time units. Equilibrium properties were calculated over the final 500 time units of each simulation. Species were considered extinct and removed when their biomass fell below 10−6. At equilibrium, both extinct species and disconnected species were removed, yielding the realised post-simulation network.

1.4 Post-simulation analyses

Post-simulation analyses quantified two complementary outcomes: community dynamics and realise network topology at equilibrium.

For community dynamics, we focused on persistence (proportion of species surviving), total community biomass, maximum trophic level, diversity (Shannon, evenness), and local stability metrics (resilience and reactivity). Local stability metrics were calculated using EcoNetPostProcessing.jl.

To assess how dynamics reshape network topology, we recalculated all structural metrics on the realised (post-extinction) network topology, enabling direct comparison between initial and emergent structure across models.

1.5 Experimental design

The analysis was designed to isolate the effect of network-generating assumptions by holding constant (i) species richness, (ii) trait distributions (within trait-based models), and (iii) dynamical rules. Differences in structure and dynamics can therefore be attributed directly to the generative model used to reconstruct trophic interactions.

2 Results

Structural Signatures: Report how much variance is explained by the model choice.

The Niche Model Template: Identify if the structural models occupy a central 'mean space' or if the dynamic models diverge into non-overlapping regions.

[Figure 1 about here.]

Divergent Stories in Dynamics: Show how/if models with similar global structures (like connectance) can lead to wildly different pathways of change/collapse/straight lines.

[Figure 2 about here.]

3 Discussion

Reconstruction is Not Neutral: Discuss how models act as structural priors that condition ecological inference. Well maybe - paleo webs seems to suggest this

Template Dependence: Address the core question: Is everything a derivative of the niche model? If dynamic models show unique cascade patterns not captured by the niche model, argue that the niche template may be insufficient for dynamic questions.

Scale-Dependent Robustness: Reflect on why species-level patterns might look similar across models while the fine-grained story of dynamic change is highly model-contingent.

Implications for Theory: Suggest that researchers must align their reconstruction framework with their inferential goals rather than defaulting to a one-size-fits-all niche template.

References

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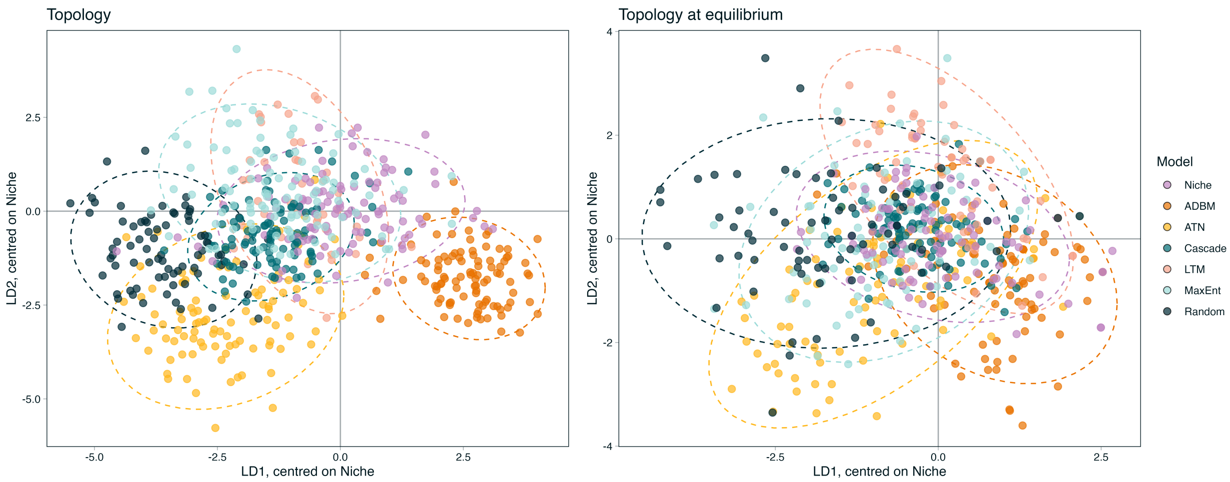

Figure 1: Linear discriminant analysis (LDA) of ecological network metrics for the seven model types. Each point represents a replicate, and ellipses indicate 95% confidence regions for each model. The second column represents the correlation of the various network metrics with the respective LDA axes. Left is before putting into END and right is post simulation structure (equilibrium topology)

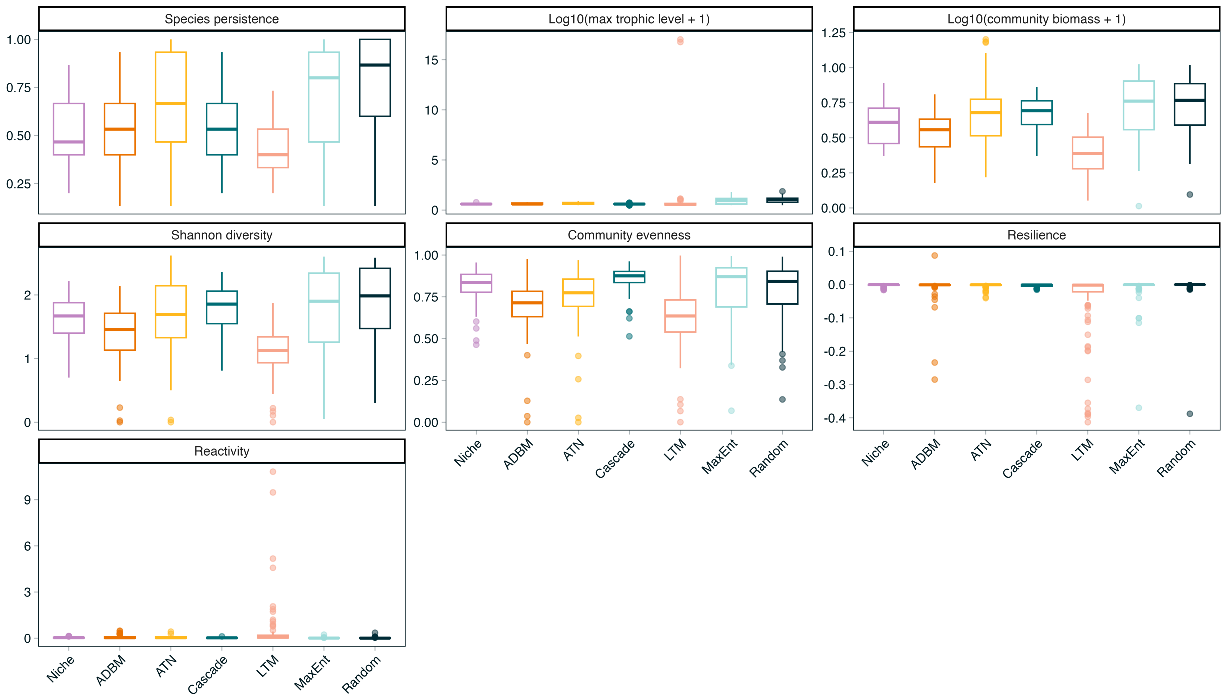

Figure 2: Boxplots showing stability metrics/information about dynamics of the system.